Chapter 12

Removal of Particles by Chemical Cleaning

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Chapter Outline

12.1 INTRODUCTION

Advanced technology nodes (65 nm and smaller) require unprecedented particle and material loss control to enable state-of-the-art device reliability and performance as indicated in the 2005 International Technology Roadmap for Semiconductors (ITRS).[1] Despite rapid decrease in device geometries in the last decade, very little has changed in surface preparation and cleaning chemistries since the introduction of the RCA clean in 1970.[2] The RCA clean used for front-end-of-line (FEOL) cleaning processes consists of two process steps known as standard clean 1 (SC-1) and standard clean 2 (SC-2). These chemistries have traditionally been used in conjunction with megasonics in an immersion bath. However, beginning with the 0.25 μm technology node it became apparent that the use of standard megasonic processes could cause pattern damage. One solution to this problem was to turn off the megasonics and increase the amount of substrate etching to undercut particles to facilitate release. However, current state-of-the-art devices are beginning to employ ultra-thin gate oxides and, therefore, material loss needs to be minimized.

In addition to device performance, there is also a market requirement for shorter manufacturing cycle times to enable "supply-on-demand"

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manufacturing. These technical and economical challenges have driven the process development of chemistries which maintain high particle removal efficiencies with reduced material loss, pattern damage and cycle time. The focus of this chapter is to highlight general surface cleaning principles to serve as a foundation for the development of new processes to meet current state-of-the-art process requirements.

Table 12.1 shows a selected list of the 2005 ITRS surface preparation requirements for FEOL processing through the 40 nm technology node.[1] In particular, at the 65 nm node the material loss target for silicon and silicon oxide is less than 0.5 Å per cleaning step while minimizing particle adders ($\geq$31.8 nm diameter) to 75. This requirement of decreasing oxide loss while maintaining high particle removal efficiency for decreasing particle sizes is currently one of the most difficult challenges in surface preparation and cleaning.

A typical wafer cleaning process consists of sulfuric acid/hydrogen peroxide/deionized water (Piranha or SPM) to remove organics. The native silicon oxide is then removed using hydrofluoric acid/deionized water [HF or dilute HF (dHF)]. Removal of particles and metals is then accomplished by ammonium hydroxide/hydrogen peroxide/deionized water (SC-1 or APM) and hydrochloric acid/hydrogen peroxide/deionized water (SC-2 or HPM). This four-step process sequence for wafer cleaning applications is known as the "B Clean." In some cases, the SC-2 may be eliminated for cleans which are not pre-diffusion. In addition, researchers have investigated the use of chelating agents which bond metal cations and solubilize them in solution where they can be rinsed easily.[3] The focus of this chapter is on particle removal which is primarily accomplished in the HF and SC-1 cleaning steps. Therefore, these processes will be reviewed in some detail along with particle removal mechanisms.

Spray processors are one type of capital equipment used in non-megasonic particle removal. A schematic of a typical batch spray processor is shown in Figure 12.1. The spray processor consists of wafer carriers mounted on a turntable that rotates about a center spray post. The rotating wafers are subjected to centrifugal force for enhanced chemical dispense uniformity and particle removal. Material loss control ($\pm 2\%$ 1σ) is achieved via a reaction rate algorithm with inputs for chemical flow and temperature. The process chamber maintains a controlled nitrogen environment to minimize chemical degradation.

In addition to batch processors, single wafer cleaning systems are also being developed for particle removal. One of the key challenges in single wafer processing is developing chemical formulations and/or chemical additives (e.g., surfactants) which improve cleaning efficiency.[4] These efficiency improvements are necessary to decrease the process time required and, thereby, increase single wafer system throughput to levels competitive with batch cleaning systems.

TABLE 12.1 2005 ITRS Selected Surface Preparation Technology Requirements[1]

	Year of Production						
	2005	**2006**	**2007**	**2008**	**2009**	**2010**	**2011**
DRAM 1/2 pitch (nm)	80	70	65	57	50	45	40
MPU/ASIC 1/2 pitch (nm)	90	78	68	59	52	45	40
MPU physical gate length (nm)	32	28	25	23	20	18	16
Wafer diameter (mm)	300	300	300	300	300	300	300
Wafer edge exclusion (mm) front surface particles	2	2	1.5	1.5	1.5	1.5	1.5
Critical particle diameter (nm)	40.1	35.7	31.8	28.4	25.3	22.5	20.1
Critical particle count (No./wafer)	94.2	59.3	75.2	94.8	59.7	75.2	94.8
Critical GOI surface metals ($\times 10^{10}$ atoms/cm^2)	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Mobile ions ($\times 10^{10}$ atoms/cm^2)	1.9	1.9	2	2.2	2.4	2.5	2.3
Surface carbon ($\times 10^{13}$ atoms/cm^2)	1.4	1.3	1.2	1	0.9	0.9	0.9
Surface oxygen ($\times 10^{13}$ atoms/cm^2)	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Surface roughness, RMS (Å)	4	4	4	4	4	2	2
Silicon loss (Å) per cleaning step	0.8	0.7	0.5	0.4	0.4	0.3	0.3
Oxide loss (Å) per cleaning step	0.8	0.7	0.5	0.4	0.4	0.3	0.3

12.2 PARTICLE/SURFACE INTERACTIONS

12.2.1 van der Waals Force

The three main interactions of interest in particle removal using wet chemicals are (1) particle–substrate (van der Waals), (2) particle–solution (electrostatic), and (3) system flow dynamics (hydrodynamic).

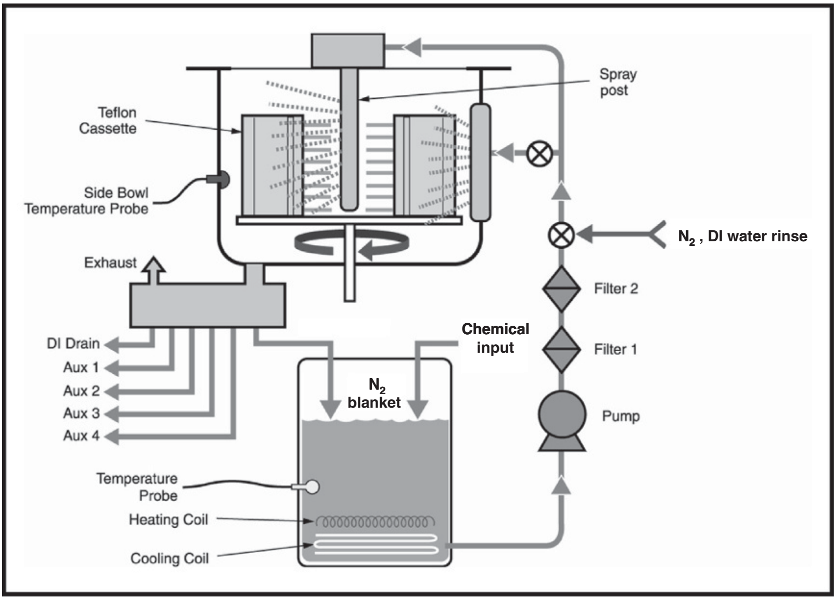

FIGURE 12.1 Schematic of a 50 wafer batch spray processor utilizing single pass or re-circulated chemical dispense. *(Source: FSI International Inc.)*

Particle–surface bonding is typically classified as physisorption or chemisorption. Physisorption is generally dominated by van der Waals or intermolecular interactions between the particle and surface. Whereas, chemisorption occurs when there is chemical bond formation (e.g., covalent or ionic) between the particle and surface atoms. The van der Waals forces are much weaker than covalent or ionic bonds. In general, newly deposited particles can be described as physisorbed. However, as the particles are "aged" over the course of days, chemical bond formation is possible and the interaction would then be described as chemisorption.

The adhesion force (F_A) of a particle physisorbed on a surface can be described via the DLVO theory as

$$F_A = F_{vdW} + F_E \tag{12.1}$$

where F_{vdW} is the van der Waals force and F_E is the electrostatic double layer force.[5] For a smooth particle on a smooth surface (Figure 12.3) the van der Waals force can be written as

$$F_{vdW} = Ad/12h^2\left(1 + 2a^2/hd\right) \tag{12.2}$$

where A is the system Hamaker constant, d is the particle diameter, h is the particle–surface separation distance, and a is the contact radius of the particle. The van der Waals force is directly proportional to the system Hamaker constant;

therefore, for systems with similar particle distributions and surface morphology the system dependent Hamaker constant can be used to qualitatively compare the van der Waals forces present in different chemical systems.[6,7] The system Hamaker constant can be written as

$$A = A_{132} = \left(\sqrt{A_{11}} - \sqrt{A_{33}}\right)\left(\sqrt{A_{22}} - \sqrt{A_{33}}\right) \tag{12.3}$$

where A_{xx} is the Hamaker constant for a particle (A_{11}) a surface (A_{22}), and the liquid (A_{33}). The calculated system Hamaker constants for common surface contaminants in silicon wafer processing are listed in Table 12.2.[8] In general, for the most commonly studied particles (Si_3N_4, SiO_2, and Si), the calculated system Hamaker constants are larger on Si surfaces when compared to SiO_2.

The Hamaker constant is useful in describing van der Waals interactions. However, studies have shown that depending on the particle composition, method of deposition, relative humidity and time since deposition, it is possible for the particle and surface interaction to be more accurately described as chemical bonding. For example, Busnaina and coworkers[9] have shown that the adhesion force of silica slurry particles deposited on silicon oxide is dependent on the relative humidity. Specifically, increasing the relative humidity increases the adhesion force to a point where covalent bond formation is believed to occur for wet-deposited particles aged on the order of several hours. In addition, depositing colloidal Si_3N_4 in an immersion bath and then processing the wafers using SPM followed by an APM that removes 3 Å of oxide leads to decreasing particle removal efficiency as the wafers age over a period of five days (Figure 12.2).

TABLE 12.2 **Hamaker Constants for Typical Particles on SiO_2 and Si Surfaces**[8]

	A_{132} ($\times 10^{-20}$ J)	
Particle	**SiO_2 Surface**	**Si Surface**
Si_3N_4	1.6	6.7
SiO_2	0.34	1.8
Al_2O_3	1.07	5.7
PSL (polystyrene latex)	0.39	1.97
Teflon	0.015	0.08
Si	1.8	9.9
Metals	1.8–2.5	10–14

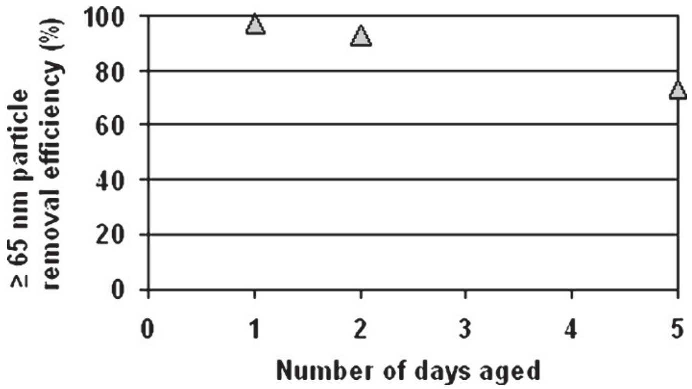

FIGURE 12.2 Particle removal efficiency versus aging for colloidal Si_3N_4 deposited from an immersion bath.

12.2.2 Electrostatic Force

Particles in solution typically acquire a surface charge arising from selective ion adsorption and/or electron exchange at the particle–liquid interface (Figure 12.3). Ions in the liquid of opposite charge will migrate to the surface of the particle via an electrostatic interaction to maintain local charge neutrality. This arrangement results in an electronic double layer at the particle–liquid interface. The thickness of the liquid double layer depends on the magnitude of the surface charge and the ionic concentration of the liquid, such that solutions with high ionic concentrations require less liquid to balance the surface charge. As the distance from the particle increases the ionic concentration decreases, eventually reaching the solution equilibrium value. The region between the highly charged region and diffuse area is known as the shear plane

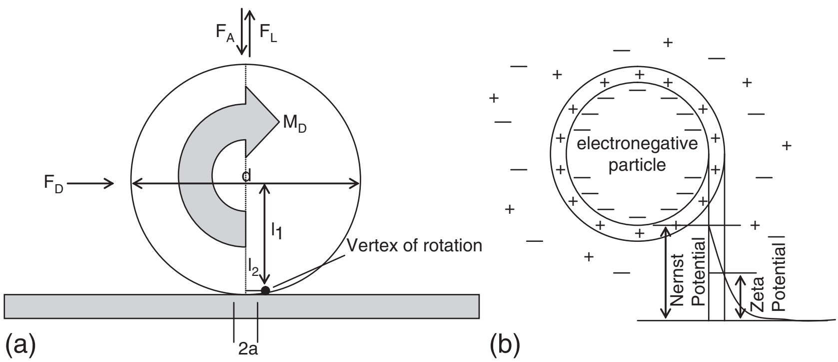

FIGURE 12.3 Schematic of (a) particle removal forces and (b) electrostatic forces. In some systems, particles can be physically removed from the surface via a "rolling mechanism" and the released particles do not redeposit on the wafer surface due to electrostatic repulsion.

TABLE 12.3 Isoelectric Points for Some Typical Particulates[8]

Particle Composition	Isoelectric Point (pH)
Si_3N_4	8.5–9.0
SiO_2	2.5–3.0
Al_2O_3	8.0–8.5
Si	2.0–2.5
Ti	6.0–6.5
TiO_2	2.5–3.0
W	2.0–2.5

and the potential at this location is known as the zeta potential. As mentioned previously, the thickness of the charged layer and, therefore, the shear plane potential is dependent on ionic concentration. In aqueous systems, the zeta potential depends on the solution pH, specifically as the pH increases (decreases) the zeta potential becomes more negative (positive). The pH which corresponds to the crossover of the zeta potential from positive to negative is known as the isoelectric point and this varies depending on particulate chemical composition. The isoelectric points for common particulates in solution are shown in Table 12.3.

The particle/particle interaction for similar particle compositions is repulsive, which leads to particle stability and limited agglomeration. If the particle charge is of opposite polarity to the wafer surface then an attractive interaction can lead to particle deposition, whereas like charges between particle and substrate are repulsive. The range of this attractive/repulsive interaction is characterized by the Debye-Hückel double layer thickness which is a measure of the radius of interaction. Vos et al.[5] have shown that for SC-1 mixtures the electrostatic interaction range increases with dilution, as a result of a decrease in ionic concentration. They have hypothesized that the standard dilutions of SC-1 (NH_4OH:H_2O_2:H_2O = 1:1:5 or 1:1:50) lead not to electrostatic double layer repulsion, but to a reduction in the range of electrostatic attraction. Hydration forces would then dominate the particle/surface interaction and are repulsive.

12.2.3 Hydrodynamic Force

Particle removal is strongly influenced by the hydrodynamic flow near the attached particle. Figure 12.4 shows particle removal efficiency (PRE) data versus oxide loss for an SC-1 process comparing batch spray to batch immersion

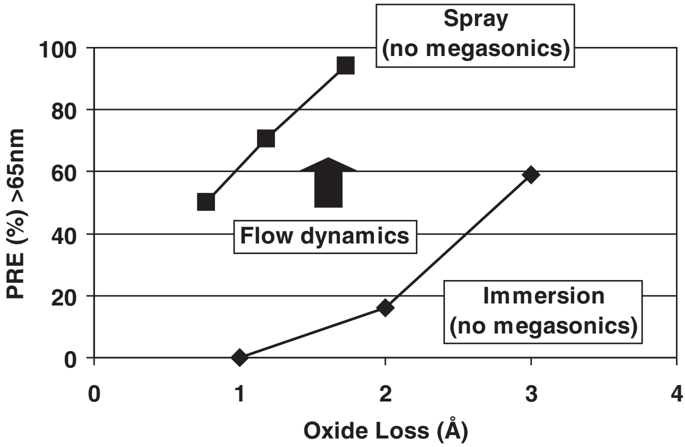

FIGURE 12.4 The >65 nm diameter particle removal efficiency (PRE) versus oxide loss for an SC-1 process comparing spray and immersion cleaning techniques.

processing. The spray process system, shown schematically in Figure 12.1, employs wafer carrier rotation about a center axis to generate a centrifugal force on the wafer surface. Immersion systems typically have considerably less hydrodynamic force across the wafer surface and rely on megasonic agitation for particle removal. In applications where megasonic damage may be a concern, the increased flow dynamics across the wafer surface in the spray processor can yield increased particle removal efficiency for a given oxide loss.

Burdick et al.,[6] using a Reynolds number approach, have developed a model to determine the critical hydrodynamic force to remove a particle physisorbed on a surface. In this model a particle Reynolds number ($\mathrm{Re_p}$) is defined as

$$\mathrm{Re_p} = d\rho V_\mathrm{p}/\mu \tag{12.4}$$

where d is the particle diameter, ρ is the fluid density, V_p is the relative velocity between the fluid and the particle center, and μ is the fluid viscosity. The hydrodynamic force to remove the attached particle depends on the removal mechanism (i.e., lifting, sliding, or rolling). Removal occurs at or above a critical Reynolds value ($\mathrm{Re_{pc}}$).

$$\mathrm{Re_p}(\text{flow}) \geq \mathrm{Re_{pc}}(\text{mechanism}) \tag{12.5}$$

Theoretical and experimental analyses indicate that the rolling mechanism has the smallest $\mathrm{Re_{pc}}$.[7] Particle morphology and surface roughness will alter the rolling point and impact the overall adhesion and hydrodynamic removal forces. However, for smooth surfaces (Figure 12.3) rolling is achieved when

$$M_\mathrm{D} + F_\mathrm{D} l_1 + F_\mathrm{L} l_2 \geq F_\mathrm{A} l_2 \tag{12.6}$$

where M_D is the external moment of surface stresses, F_D is the drag force, l_1 is the vertical lever arm, F_L is the lift force, l_2 is the horizontal lever arm, and F_A is the adhesion force.

As mentioned previously, chemisorbed particles are bound much more strongly to the surface compared to physisorbed particles. Hydrodynamic forces alone result in limited removal of particles chemisorbed to surfaces. In these cases, bond dissociation reactions are required to lower the adhesion force to allow the particles to be removed by hydrodynamic force.

In practice, particle removal is achieved through multiple mechanisms including particle undercutting via surface etching, particle removal via a rolling mechanism and electrostatic double layer repulsion to prevent released particles from re-depositing on the wafer surface, all of which are required to achieve high particle removal efficiencies. Particle removal becomes more difficult as particle size decreases because the adhesion energy is typically proportional to the particle contact area, whereas the removal forces typically vary with the particle volume. Consequently, as particle size decreases the removal force decreases at a faster rate compared to the adhesion force.[5]

12.3 PROCESS APPLICATIONS AND CHEMISTRIES

12.3.1 Particle Challenge Wafer Preparation

Particle removal efficiency is strongly dependent on challenge wafer preparation including method of particle deposition (wet-dipped or aerosol), particle composition and particle size distribution. At the present time, organizations which provide industry guidance (e.g., SEMI and ITRS) have not specified a guideline relating to particle removal challenge preparation.

Figure 12.5 shows particle removal efficiency using an SPM-APM process with increasing oxide removal. The particles were prepared and then aged for 24 h. The "wet" particle challenge wafers were prepared by placing

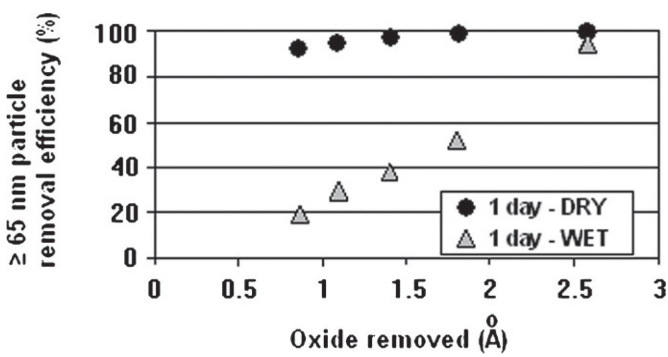

FIGURE 12.5 Particle removal efficiency versus oxide loss for colloidal Si_3N_4 deposited via dry and wet-deposition methods.

LIVE GRAPH
Click here to view

polycrystalline Si_3N_4 into an immersion bath containing silicon wafers. The "dry" particle challenge wafers were prepared using the same colloidal Si_3N_4 in a commercial aerosol deposition system (MSP Corporation, Shoreview, MN). The data clearly show that higher particle removal efficiency versus oxide loss is observed for the dry deposited particles. Only 1 Å oxide loss was needed to remove >90% of the dry deposited Si_3N_4 particles as compared to the 2.5 Å needed to remove >90% of the wet deposited Si_3N_4 particles.

12.3.2 dHF Clean

Vos et al.[8] examined the use of dilute HF (dHF) mixtures for inorganic and organic removal efficiencies on both silicon oxide and silicon surfaces. Here, only inorganic removal is considered because typical wafer clean applications will utilize SPM or DI Ozone for organic removal prior to particle removal processing. Their data indicate that for particles on thermal oxide surfaces the adhesion energy is dominated by electrostatic interactions. Specifically, for a dHF solution of pH 1.9, particle removal was very high (>90%) for SiO_2, TiO_2, Ti, and W, whereas, it was lower for Si_3N_4, Al_2O_3, and Si. Considering the isoelectric points for the particles (Table 12.3), one would expect strong electrostatic attraction to the thermal silicon oxide surface for Si_3N_4, Al_2O_3, and Ti. Ti, however, is easily removed from SiO_2 which is readily explained based on known HF etching of Ti.[8] On silicon surfaces, particle removal was very high (>90%) for Si_3N_4, Al_2O_3, SiO_2, TiO_2, and Ti, whereas it was lower for Si and W. Since the SiO_2 and Si surfaces have very similar zeta potentials at pH = 1.9, one would expect similar particle removal trends. Therefore, electrostatic attraction does not adequately explain the observed phenomena on silicon surfaces. Considering the Hamaker constants for these particle-substrate compositions, the Si surfaces tend to have much stronger van der Waals interactions compared to the SiO_2 surfaces. Consequently, particle removal on silicon surfaces is dominated by van der Waals interactions and not electrostatic. The lower particle removal for Si and W on silicon surfaces is consistent with their relatively high Hamaker constants (Table 12.2). Once again, the trend deviates for Ti due to removal via HF etching.

The use of anionic surfactants in dHF solutions of pH 1.9 and higher can improve particle removal efficiency for positively charged particles (e.g., Si_3N_4, Al_2O_3) by reversing the zeta potential.[8] This occurs through selective adsorption of the anionic head group of a long hydrocarbon chain surfactant. The modified particle will then be electrostatically repelled from the negatively charged SiO_2 or Si wafer surface.

12.3.3 SC-1 Clean

The SC-1 (also known as ammonia/peroxide mixture or APM) process comprises an aqueous mixture of NH_4OH:H_2O_2:H_2O. The SC-1 process mechanism involves oxidizing the silicon surface using H_2O_2 and simultaneous removal of the silicon oxide using NH_4OH at alkaline pH. Much work has been done on optimizing this ratio to reduce chemical consumption and material loss, as well as increasing particle removal efficiency. Although the original ratio of $NH_4OH:H_2O_2:H_2O$ is 1:1:5, it is now quite common for dilutions of 1:1:50–1:1:500 to be used in production. The dilution does not significantly affect the etch rate of silicon oxide as shown in Figure 12.6.

Alternatively, the $NH_4OH:H_2O_2$ ratio is important in material loss, specifically silicon loss. Figure 12.7 shows the silicon and silicon oxide losses versus $NH_4OH:H_2O_2$ ratio. The silicon oxide loss remains relatively constant over the range of 1:2–1:10, however, the silicon loss decreases ~30% over this range.

Although dilution of SC-1 does not change material loss, the resultant decrease in solution ionic strength and NH_4^+ concentration can alter metal

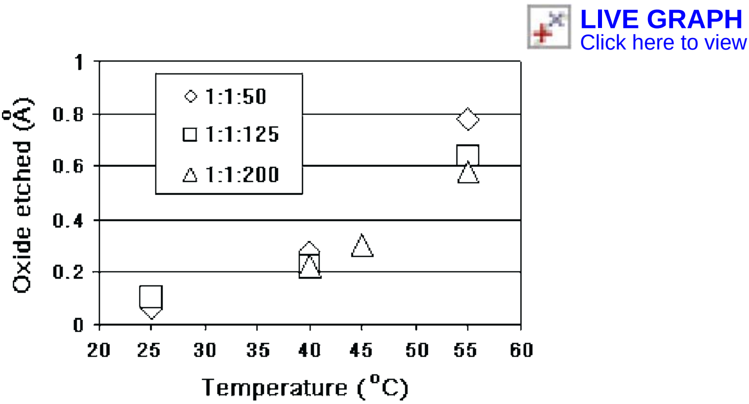

FIGURE 12.6 The dependence of oxide loss on SC-1 temperature and SC-1 dilution. The oxide loss is only weakly dependent on dilution, however, it is a strong function of temperature in the range of parameters examined.

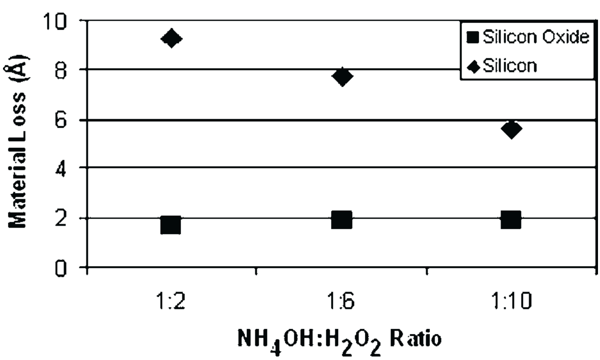

FIGURE 12.7 Material loss versus APM ratio for silicon oxide and silicon. Increasing the H_2O_2 over the range shown results in decreasing silicon loss but a constant silicon oxide loss.

contamination levels. Loewenstein and Mertens[10] have examined the effect of SC-1 concentration on surface metal contamination. They found that NH_3 and the conjugate acid, NH_4^+, could act as a complexing agent and a competing ion adsorber, respectively. As a result, if a metal cation forms a thermally stable amine complex (e.g., Ni^{2+} or Zn^{2+}) then surface levels of these metals will be reduced with increasing NH_4OH concentration. For metal cations which do not form amine complexes (e.g., Ca^{2+}) then NH_4^+ cations can compete for surface adsorption. These authors calculate that the NH_4^+ cation adsorbs at a rate comparable to or greater than H^+ which is significantly higher than the rate at which Ca^{2+} adsorbs, therefore, NH_4^+ is thermodynamically favored over Ca^{2+}. In summary, very dilute SC-1 mixtures may lead to similar particle removal efficiencies for a given silicon oxide loss, however, an increase in metal contamination may be observed.

As discussed previously, silicon etching is very sensitive to the NH_4OH:H_2O_2 ratio, such that, as the H_2O_2 decreases the silicon etch rate increases. The starting SC-1 ratio may change throughout the course of wafer processing via three mechanisms, (1) NH_3 evaporation followed by condensation on the wafer surface, (2) metal contamination present in the SC-1 solution (e.g., Fe) which is insoluble in the SC-1 mixture and can catalyze the decomposition of H_2O_2, and (3) metal contamination present in upstream processing (e.g., Cu in dHF mixtures) which can decompose H_2O_2 via a galvanic oxidation/reduction reaction.[11,12] These decomposition mechanisms are particularly important where batch immersion or recirculation chemistries are employed. In a typical batch immersion system during transition into and out of the SC-1 tank, the wafers can be exposed to NH_3 vapor which can condense on the wafer. This NH_3-rich solution can then selectively etch the silicon surface along the [111] plane.[12] Depending on the method of dispensing and chemical temperature, evaporation could significantly affect reactive agent concentration and the APM ratio; as a result, bath monitoring is critical to ensure process repeatability.

Metal incorporation into the chemically grown SC-1 oxide can also be a significant issue in achieving adequate device performance. Knotter et al.[12] exposed hydrophilic and hydrophobic surfaces to a 1 wt ppb Fe-contaminated APM bath followed by drying in a nitrogen environment. The Fe levels on the wafer surface were then measured using TXRF (total reflection X-ray fluorescence). The hydrophilic wafer Fe levels were 11.7×10^{10} and 19.0×10^{10} atoms/cm^2 for the hydrophobic wafer. The hydrophobic surfaces were then rinsed in deionized water followed by drying in a nitrogen environment. The Fe level decreased to 12.3×10^{10} atoms/cm^2 consistent with surface Fe being readily removed with remaining Fe atoms incorporated into the oxide layer. Metal incorporation into the chemical grown oxide is one reason that an "HF-last" process is required to selectively remove silicon oxide prior to high-temperature diffusion processing. Although particle removal is the focus of this chapter, one needs to be cognizant of the potential effects on metal contamination during particle removal process steps.

12.3.4 Single-Step Clean

While the two-step RCA clean has been the industry standard since 1970, much effort has been spent on reducing it to a single step. The benefits of single-step processing include: (1) reduced tank requirements for immersion systems, (2) cycle-time reduction, and (3) reduced DI water consumption from the elimination of intermediate rinse steps.[13]

The SC-2 (also known as hydrochloric/peroxide mixture or HPM) process utilizes a mixture of $HCl:H_2O_2:H_2O$, traditionally in a 1:1:6 ratio. The SC-2 process mechanism involves solubilizing metal ions and metal hydroxides via the reaction

$$-\mathrm{Si}-\mathrm{O}-\mathrm{H}+\mathrm{M}^{x+} \leftrightarrow -\mathrm{Si}-\mathrm{O}-\mathrm{M}^{(x-1)}+\mathrm{H}^{+} \qquad (12.7)$$

In this reaction, metal is removed from the silicon surface by adding an acid (e.g., HCl) which generates a metal cation in solution which, in turn, can form an ionic bond with the chloride anion and be rinsed away.[14] The metal-ion contaminants originate mainly from the chemicals used in the SC-1 process with a pH range where most metal species are insoluble. However, commercial sources of SC-1 chemical constituents have improved since the development of the original RCA process. As a result, the SC-2 process has been eliminated in many wafer cleaning applications, particularly for processes which are not pre-diffusion cleans. In cases where metal complexing is necessary, chelating agents can be added directly to the SC-1 process. These chelating agents can bind the metal species in aqueous solution and be rinsed away.[3,15] In addition, surfactants can be used to augment the electrostatic repulsion of the contaminants to improve particle removal efficiency.[16] However, care must be taken that these chemical additives do not irreversibly adsorb onto the silicon substrate and create interface adhesion and/or reliability problems.

12.4 SUMMARY

In conclusion, particle removal is strongly dependent on the particle interactions with the substrate and surrounding liquid. Depending on the solution pH, substrate composition and particle aging, the intermolecular interactions are governed to varying degrees by van der Waals, covalent, and electrostatic forces. Industry standards are needed to accurately compare particle removal techniques; for the particle challenge wafers examined here, $>90\%$ particle removal efficiency required 1 and 2.5 Å silicon oxide losses for dry- and wet-deposited Si_3N_4 particles, respectively. Material loss requirements for the 45 nm technology node as indicated in the 2005 ITRS roadmap suggest that further process development is required to maintain high particle removal efficiency with $\leq$0.3 Å material loss per cleaning step without the use of megasonics. Current market drivers require that process chemistries address both particle and metal removal within the same process step without degrading electrical device performance through irreversible chemical additive adsorption. The process mixture physical and chemical properties (e.g., pH, solubility, etch selectivity) and the method of dispensing (e.g., spray, immersion) are critical factors in optimizing particle removal for advanced semiconductor device manufacturing.

REFERENCES

1. The International Technology Roadmap for Semiconductors is available online at http://www.itrs.net/.
2. Kern, W. Overview and Evolution of Semiconductor Wafer Contamination and Cleaning Technology. In: *Handbook of Semiconductor Wafer Cleaning Technology: Science Technology, and Applications;* Kern, W., Ed.; Noyes Publications: Park Ridge, NJ, 1993; pp 3–67.
3. Gale, G. W.; Rath, D. L.; Cooper, E. I.; Estes, S.; Okorn-Schmidt, H. F.; Brigante, J.; Jagannathan, R.; Settembre, G.; Adams, E. Enhancement of Semiconductor Wafer Cleaning by Chelating Agent Addition. *J. Electrochem. Soc.* **2001**, *148*, G513.
4. Baker, J.; Beaudry, C.; Morinaga, H.; Verhaverbeke, S. Surfactant Selection for AM Clean in Single Wafer Oasis Wet System. *Diffusion and Defect Data Pt. B: Solid State Phenomena* **2003**, *92*, 57.
5. Vos, R.; Xu, K.; Vereecke, G.; Holsteyns, F.; Fyen, W.; Wang, L.; Lauerhaas, J.; Hoffman, M.; Hackett, T.; Mertens, P.; Heyns, M. Advanced Wet Cleaning of Sub-micrometer Sized Particles. In: *Particles on Surfaces 8: Detection, Adhesion and Removal;* Mittal, K. L., Ed.; VSP: Utrecht, The Netherlands, 2003; pp 255–270.
6. Burdick, G. M.; Berman, N. S.; Beaudoin, S. P. Describing Hydrodynamic Particle Removal from Surfaces using the Particle Reynolds Number. *J. Nanoparticle Res.* **2001**, *3*, 455.
7. Burdick, G. M.; Berman, N. S.; Beaudoin, S. P. A Theoretical Evaluation of Hydrodynamic and Brush Contact Effects on Particle Removal during Brush Scrubbing. *J. Electrochem. Soc.* **2003**, *150*, G658.
8. Vos, R.; Lux, M.; Xu, K.; Fyen, W.; Kenens, C.; Conard, T.; Mertens, P. W.; Heyns, M. M.; Hatcher, Z.; Hoffman, M. Removal of Submicrometer Particles from Silicon Wafer Surfaces using HF-Based Cleaning Mixtures. *J. Electrochem. Soc.* **2001**, *148*, G683.
9. Adams, G.; Busnaina, A.; Muftu, S. *Particle Adhesion and Removal in Post-CMP Applications*; Microcontamination Research Lab, Northeastern University: Boston, MA, 2002. Presentation available at www.cmc.neu.edu.
10. Loewenstein, L. M.; Mertens, P. W. Competitive Adsorption of Cations onto the Silicon Surface: The Role of the Ammonium Ion in Ammonia-Peroxide Solution. *J. Electrochem. Soc.* **1999**, *146*, 3886.
11. Knotter, D. M.; de Gendt, S.; Baeyens, M.; Mertens, P. W.; Heyns, M. M. Hydrogen Peroxide Decomposition in Ammonia Solutions. *J. Electrochem. Soc.* **1999**, *146*, 3476.
12. Knotter, D. M.; de Gendt, S.; Mertens, P. W.; Heyns, M. M. Silicon Surface Roughening Mechanisms in Ammonia Hydrogen Peroxide Mixtures. *J. Electrochem. Soc.* **2000**, *147*, 736.
13. Chao, T. S.; Yeh, C. H.; Pan, T. M.; Lei, T. F.; Li, Y. H. A One-Step Single-Cleaning Solution for CMOS Processes. *J. Electrochem. Soc.* **2003**, *150*, G503.
14. Verhaverbeke, S.; Kuppurao, S.; Beaudry, C.; Truman, J. K. Single-Wafer, Short Cycle Time Wet Clean Technology. *Semiconductor Intl.* **2002**, *25*(July), 91.

15. Mertens, P. W.; Parton, E. Sub-100 nm Technologies Drive Single-Wafer Wet Cleaning. *Solid State Technol*. **2002**, *45*(February), 51.
16. Free, M. L. The Use of Surfactants to Enhance Particle Removal from Surfaces. In: *Developments in Surface Contamination and Cleaning;* Kohli, R.; Mittal, K. L., Eds.; William Andrew Publishing: Norwich, NY, 2008; pp 727–758.